

Lead-lag timing is now directly testable

RULING - DEMOTE TO EXPLORATORY. Exact timing is now verifiable, but the quality-screened pairs do not satisfy every frozen confirmatory-support criterion.

149	110	41	37.3%
Fmask files hash-verified and opened	candidate pass-window-sensor records	verified quality-screened pre/post pairs	overall paired share; ruling uses frozen strata

Quality screen

Each HLS V2 Fmask was clipped to the frozen Census urban domain and the non-overlapping MGRS core. A cell survived when source data were present, Fmask bits 1-5 were zero, and aerosol bits 6-7 were no greater than two. A 1 km acquisition-block required at least 60 surviving 30 m cells.

Pair construction

For every thermal pass and sensor, the analysis required distinct source identifiers: one HLS acquisition 1-15 days before and another 1-15 days after, with absolute lags within three days. A pair also needed at least 30 QA-usable study blocks in common. The frozen lag and stable-ID tie-breaker was applied after QA.

What changed from D010

D010 could not estimate timing because the inherited centered Sentinel-2 object lacked source dates and identifiers. The D018 reassessment uses exact HL30.002 and HLSS30.002 metadata plus the acquired Fmask files, so pre/post direction, distinct identity, lag balance, spatial QA support and acquisition reuse are now measurable.

Interpretation boundary

This remains a feasibility count, not an exposure or outcome analysis. No HLS reflectance, optical index, ECOSTRESS LST, or v6.2 coefficient was opened. Passing this check would not authorize Gate A. The final antecedent and future-placebo variables must still be rebuilt separately from raw HLS reflectance after authorization.

Support by city and season window

Aggregated timing potential

City	Window	Candidates	Verified pairs	Share
Phoenix	Primary	10	5	50.0%
Phoenix	Sensitivity	34	15	44.1%
Los Angeles	Primary	50	15	30.0%
Los Angeles	Sensitivity	16	6	37.5%

The formal ruling is not based on these pooled shares. It applies the frozen 70% criterion to every nonzero city-year-window-sensor stratum and checks acquisition reuse.

Verified lag distribution

Pre median 8.16 d	Pre p90 13.01 d	Post median 8.71 d	Post p90 13.83 d
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Formal ruling

The fail-closed result is **DEMOTE_TO_EXPLORATORY_CONFIRMATORY_SUPPORT_RULE_FAILED**. Neither city has a frozen window meeting every nonzero-stratum pair-share and acquisition-reuse limit. Timing provenance is repaired, but confirmatory support is not.

Limitations carried forward

The 149 rasters came from the catalogue-selected pair plan. A missing pair is therefore conservative with respect to alternative HLS acquisitions whose Fmask was not downloaded. The screen is also optimistic because it does not yet require actual tree/background pixels, reflectance validity, reliability, or the final exposure construction.

Files to inspect

The 56-row table reports all required strata. The timing figure reports every selected pre/post lag. The JSON evidence retains per-acquisition QA coverage and each candidate-pair result.